

Geography Ch-13 (Advanced): India's Weather Systems - Jet Streams, Western Disturbances & Local Winds

Tier 2: Advanced | Source: D.R. Khullar (India: A Comprehensive Geography) + Majid Husain (Geography of India)

India-applied climatology grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

India-Applied Weather: Jet Streams, Western Disturbances, Loo & Regional Climatic Controls

GS-I (Physical Geography of India) Geography

■ NEWS TRIGGER

Advanced application (D.R. Khullar; Majid Husain, Geography of India). India's weather is a product of its controls - latitudinal extent (Tropic of Cancer bisects), the Himalayan wall blocking cold Central Asian air, distance from sea, relief, and the upper-air jet streams.

■ INTRO & ORIGIN

Beyond the planetary framework, India's weather is shaped by upper-tropospheric jet streams (Subtropical Westerly Jet in winter, Tropical Easterly Jet in summer), the Tibetan-Himalayan highlands, and imported extra-tropical Western Disturbances. These explain why India's winter rain, monsoon onset and heatwaves behave the way they do.

Blanford and later climatologists noted India's 'broad climatic unity' (monsoon) masks huge regional variation in wind, temperature and rainfall due to location, altitude, distance from sea and relief (D.R. Khullar).

■ KEY DATA

System	Season / Mechanism	Effect on India
Subtropical Westerly Jet (STWJ)	Winter; south of Himalayas ~25 deg N	Steers Western Disturbances into NW India; winter rain
Tropical Easterly Jet (TEJ)	Summer; over peninsular India ~15 deg N	Aids monsoon burst; linked to Tibetan Plateau heating
Western Disturbances (WD)	Mediterranean origin, moved by STWJ	Winter rain (mahawat) + Himalayan snow - vital for Rabi crops
Loo	May-June, steep NW pressure gradient	Hot dry local wind; heatwave conditions
Tibetan-Himalayan highlands	Barrier + heat source	Blocks cold air; drives monsoon reversal

■ STATIC THEORY

- **Jet streams:** fast, narrow bands of upper-tropospheric westerly wind. The **Subtropical Westerly Jet (STWJ)** sits south of the Himalayas in winter and is the 'conveyor belt' that guides Western Disturbances into NW India. Its withdrawal north of the Himalayas in summer is a pre-condition for monsoon onset.
- **Tropical Easterly Jet (TEJ):** forms in summer over peninsular India (~15 deg N) due to intense heating of the Tibetan Plateau; its establishment helps trigger the monsoon 'burst'.
- **Western Disturbances (WD):** extra-tropical (non-monsoonal) low-pressure systems originating over the Mediterranean, travelling eastward via Iran-Afghanistan-Pakistan, steered by the STWJ. They bring winter/early-spring rain to Punjab, Haryana, Delhi, UP and snow to J&K, HP, Uttarakhand - critical for Rabi (wheat, mustard, gram).
- **Loo (Khullar):** in May-June, intense heating of NW India builds a steep pressure gradient (~1.0-1.5 mb per degree latitude), generating the hot, dust-laden Loo wind that starts ~9 AM and peaks in the afternoon.
- **Regional controls (Khullar):** latitudinal extent (Tropic of Cancer), altitude (temperature falls ~6.5 deg C/km), continentality (interior extremes vs coastal moderation), and the Himalayan barrier that keeps Central Asian cold air out - giving India warmer winters than comparable latitudes.

■ MUST-KNOW FACTS

1. STWJ steers Western Disturbances in winter; its northward shift is essential for SW monsoon onset.

■ UPSC TRAPS

✗ Western Disturbances are part of the SW monsoon.

2. TEJ (Tropical Easterly Jet) is a SUMMER feature over peninsular India, unique to the monsoon system.
3. Western Disturbances = Mediterranean origin, extra-tropical, bring winter rain 'mahawat' vital for wheat.
4. Tibetan Plateau acts as an elevated heat source in summer (drives low pressure and monsoon indraft).
5. Himalayas give India a tropical-monsoon character by blocking cold Siberian air masses.

✓ They are extra-tropical, non-monsoonal systems from the Mediterranean, active in WINTER, unrelated to the SW monsoon.

✗ The Tropical Easterly Jet operates in winter.

✓ TEJ is a summer phenomenon; the Subtropical Westerly Jet dominates in winter.

✗ Loo is felt uniformly across all India.

✓ Loo is confined to the northern plains (NW India); peninsular and coastal India, moderated by the sea, do not experience it.

■ MAINS ANGLE

Analyse the role of jet streams (STWJ and TEJ) and Western Disturbances in shaping the seasonal rhythm of India's weather, and how their variability threatens Rabi-crop food security.

■ STUDY LINK

Geography -> Climate of India -> Jet Streams & Western Disturbances (leads into the Indian Monsoon mechanism, Ch-14)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: 2026 - Hyperactive Western Disturbances, Jet-Stream Shifts & El Nino Monsoon Worry

GS-I (Climate of India) + GS-III (Agriculture / Climate Change)
Geography - Current Affairs

■ NEWS TRIGGER

2026: IMD recorded a higher-than-normal number of intense Western Disturbances (April 2026 rainfall ~12% above normal) causing pre-monsoon thunderstorms/hailstorms; simultaneously a strong El Nino probability by July 2026 raised fears of a below-normal monsoon (IMD forecast ~90% of LPA). Down To Earth linked excess pre-monsoon rain to East-West anticyclones.

■ INTRO & ORIGIN

2026 showcased how climate change is destabilising India's upper-air systems: more frequent and intense Western Disturbances (steered by a wavier Subtropical Westerly Jet) delivered excess spring rain and hail, even as an emerging El Nino threatened to suppress the SW monsoon - a whipsaw with direct agricultural stakes.

IMD & climate researchers note climate change is altering jet-stream behaviour (more meandering/blocking), changing WD frequency and track, and interacting with ENSO to produce erratic, high-variance rainfall.

■ KEY DATA

2026 Phenomenon	Cause (static link)	Consequence
Hyperactive Western Disturbances	Wavier STWJ + moisture from Arabian Sea/BoB	April rain 12% above normal, hail damage to Rabi crops
East-West anticyclones	Anticyclonic moisture convergence	Excess pre-monsoon thunderstorms (Down To Earth)
El Nino developing by Jul 2026	Warm central-east Pacific SST	Suppressed monsoon; IMD ~90% of LPA forecast
Delayed/erratic monsoon onset	Jet-stream + ENSO interaction	Drought and food-security concern

■ STATIC THEORY

- **El Nino (ENSO warm phase):** abnormal warming of the central-eastern equatorial Pacific weakens the Walker Circulation, tends to suppress the Indian SW monsoon (below-normal rainfall). La Nina generally does the opposite.
- **Jet-stream waviness:** a warming Arctic reduces the equator-to-pole temperature gradient, making the polar/subtropical jets more meandering ('blocking'), which prolongs weather extremes (both heat domes and stalled rain systems).
- **LPA (Long Period Average):** IMD's benchmark for monsoon rainfall; 96-104% of LPA = 'normal'. 90% = 'below normal'.

■ MUST-KNOW FACTS

1. April 2026 all-India rainfall ~12% above normal, driven by active Western Disturbances (IMD).
2. El Nino projected by July 2026 -> IMD flagged risk of below-normal monsoon (~90% of LPA).
3. Down To Earth attributed excess pre-monsoon rain to East-West anticyclones over India.
4. Rabi crops (wheat, mustard, gram) depend on WD winter rain - excess/hail damages standing crops.
5. Normal monsoon = 96-104% of LPA (IMD classification).

■ UPSC TRAPS

- ✗ El Nino strengthens the Indian monsoon.
- ✓ El Nino typically SUPPRESSES the Indian SW monsoon (below-normal rain); La Nina tends to enhance it.
- ✗ More Western Disturbances are always good for agriculture.
- ✓ Excess/intense WDs bring hailstorms and untimely rain that damage standing Rabi crops - too much is as harmful as too little.

■ MAINS ANGLE

Discuss how climate-change-driven jet-stream variability and ENSO simultaneously threaten India with hyperactive Western Disturbances and weak monsoons - a double risk to agrarian food security.

■ STUDY LINK

Geography -> Climate of India -> Jet Streams / WD + ENSO; connect to GS-III agriculture & climate-adaptation policy